



NORTH SYDNEY BOYS HIGH SCHOOL

2010 HSC ASSESSMENT TASK 4

Mathematics

Extension 1

General Instructions

- Working time – 50 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Ireland
- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Barrett
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	Total	Total %
Mark	$\frac{\quad}{6}$	$\frac{\quad}{10}$	$\frac{\quad}{6}$	$\frac{\quad}{9}$	$\frac{\quad}{14}$	$\frac{\quad}{5}$	$\frac{\quad}{50}$	$\frac{\quad}{100}$

Question 1 (6 marks)**Marks**

A particle moves linearly so that its displacement x metres from the origin O is given by $x = \sqrt{3} \cos 3t - \sin 3t$.

- (i) Show that the equation can be expressed as $x = 2 \cos(3t + \frac{\pi}{6})$. 2
- (ii) Show that the motion is simple harmonic. 2
- (iii) Find the time at which the particle first passes through the origin. 2

Question 2 (10 marks)

When a particle moves along the x -axis with velocity v at time t , its acceleration is given by $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

The velocity $v \text{ ms}^{-1}$ of a particle moving in simple harmonic motion along the x axis is given by the expression $v^2 = 28 + 24x - 4x^2$.

- (i) Between which two points is the particle oscillating? 2
- (ii) What is the amplitude of the motion? 1
- (iii) Find the acceleration in terms of x . 2
- (iv) What is the centre of the motion? 1
- (v) Find the period of the oscillation. 2
- (vi) If the particle starts from the position furthest to the right, find the displacement function in terms of t . 2

Question 3 (6 marks)

The velocity of a particle moving in a straight line at position x is given by $v = 2e^{-x}$. The particle is initially at the origin.

- (i) Show that the acceleration at position x is given by $a = -4e^{-2x}$. 2
- (ii) What is the initial acceleration? 1
- (iii) Find the displacement function giving x at time t . 3

Question 4 (9 marks)

- (a) $(1 + \sqrt{3})^4 = a + b\sqrt{3}$, where a, b are integers. By expansion, find a and b . 3
- (b) Find the value of n such that when $(1 + x)^n$ is expanded in ascending powers of x , the coefficient of x^4 is twice the coefficient of x^3 . 3
- (c) Find the exact value of the coefficient of x^{12} in the expansion of

$$\left(2x - \frac{1}{x^2}\right)^{30} \quad 3$$

Question 5 (14 marks)

(a) Isaac calculates that he will need \$1 000 000 if he is to retire in 20 years' time and maintain his present lifestyle. He decides to invest a fixed amount $\$P$ at the beginning of each quarter in an investment account that pays interest at an annual rate of 8% compounding quarterly.

Let A_n be the amount in dollars in Isaac's investment account at the end of the n^{th} quarter.

- (i) Show that $A_2 = \$P \times (1.02 + 1.02^2)$ 2
- (ii) Find a similar expression for the value of his investment after 20 years and hence calculate the value of P needed to realise the amount of \$1 000 000 needed for his retirement. (*Show all appropriate working*). 3

(b) Mr Wu decides to borrow \$250 000 to buy a unit. Interest is calculated monthly on the balance still owing, at a rate of 6% per annum. The loan is to be repaid at the end of 15 years with equal monthly repayments of $\$M$. Let A_n be the amount owing after the n^{th} repayment.

- (i) Show that $A_{60} = 250\,000(1.005)^{60} - M(1 + 1.005 + \dots + 1.005^{59})$ 2
- (ii) Find the value of M , showing all appropriate working. 2
- (iii) Hence calculate the amount still owing after 5 years of payment at this rate. 2
- (iv) At the end of five years the interest rate is increased to 7.2% per annum and Mr Wu changes his payments to \$2400 per month. How many months are required to pay off the remainder of the loan? 3

Question 6 (5 marks)

(a) By considering the expansion of the identity $(1+x)^6 (1+x)^6 \equiv (1+x)^{12}$, show that

$$\sum_{k=0}^6 \binom{6}{k}^2 = \binom{12}{6} \quad 3$$

(b) Solve the inequality $x^4 + 4x^3 + 6x^2 + 4x + 1 \leq x^3 + 3x^2 + 3x + 1$ 2

Q1

$$\begin{aligned}
 \text{(i)} \quad x &= \sqrt{3} \cos 3t - \sin 3t \\
 &= 2 \left(\frac{\sqrt{3}}{2} \cos 3t - \frac{1}{2} \sin 3t \right) \\
 &= 2 \cos \left(3t + \alpha \right)
 \end{aligned}$$

$$\text{where } \cos \alpha = \frac{\sqrt{3}}{2}, \quad \sin \alpha = \frac{1}{2} \quad \therefore \alpha = \frac{\pi}{6}$$

$$\therefore x = 2 \cos \left(3t + \frac{\pi}{6} \right)$$

$$\text{(ii)} \quad \dot{x} = -6 \sin \left(3t + \frac{\pi}{6} \right)$$

$$\ddot{x} = -18 \cos \left(3t + \frac{\pi}{6} \right)$$

$$\text{But } x = 2 \cos \left(3t + \frac{\pi}{6} \right)$$

$$\therefore \ddot{x} = -9x$$

This is of form $\ddot{x} = -n^2 x$

$\therefore$ motion is S.H.M.

$$\text{(iii)} \quad \text{At } t=0, \quad x = 2 \cos \frac{\pi}{6} = \sqrt{3}$$

$$\text{and } v = -6 \sin \frac{\pi}{6} = -3 \text{ m/s}$$

So at $t=0$ the particle is moving towards the origin.

$$\text{At } x=0 \text{ we have } \cos \left(3t + \frac{\pi}{6} \right) = 0$$

$$\therefore 3t + \frac{\pi}{6} = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$\therefore t = \frac{\pi}{9}, \frac{4\pi}{9}, \dots$$

So first through origin at $t = \frac{\pi}{9}$ secs.

✓ Derives amplitude
(Also: $a = \sqrt{3+1} = 2$.)

✓ Derives auxiliary angle correctly.
(showing in working that it is in 1st quadrant)

✓
✓

✓

✓

Q2

$$\begin{aligned}
 \text{(i)} \quad v^2 &= 28 + 24x - 4x^2 \\
 &= 4(7 + 6x - x^2) \\
 &= 4(7-x)(1+x)
 \end{aligned}$$

Since $v^2 \geq 0$, $\therefore 4(7-x)(1+x) \geq 0$ ✓

$\therefore -1 \leq x \leq 7$ ✓

(ii) amplitude = $\frac{7 - (-1)}{2} = 4$ ✓

$$\begin{aligned}
 \text{(iii)} \quad \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\
 &= \frac{d}{dx} (14 + 12x - 2x^2) \\
 &= 12 - 4x \\
 \ddot{x} &= -4(x-3)
 \end{aligned}$$

✓

✓

(iv) Centre of motion is $x=3$ ✓

(v) $T = \frac{2\pi}{n}$

Now $\ddot{x} = -n^2(x-3) \therefore n=2$ ✓ for n

$\therefore T = \frac{2\pi}{2} = \pi$ seconds. ✓ for T

(vi) $x = 3 + 4 \cos(2t + \alpha)$

At $t=0$, $x=7$

$\therefore 7 = 3 + 4 \cos \alpha$

$\cos \alpha = 1$

$\alpha = 0$

$\therefore x = 3 + 4 \cos 2t$

✓ appropriate start

✓ or equivalent equation

3

$$v = 2e^{-x}$$

$$(i) \quad a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$= \frac{d}{dx} \left(\frac{1}{2} \cdot 4 \cdot e^{-2x} \right)$$

$$= -2 \cdot 2e^{-2x}$$

$$\therefore a = -4e^{-2x}$$

✓ correct start

✓ shows correct working

$$(ii) \quad \text{When } t=0, x=0$$

$$\therefore a = -4e^0$$

$$= -4$$

$\therefore$ initial acceleration is -4 units.

✓ for -4

$$(iii) \quad \frac{dx}{dt} = 2e^{-x}$$

$$\therefore \frac{dt}{dx} = \frac{e^x}{2}$$

$$t = \frac{1}{2}e^x + C$$

$$\text{When } t=0, x=0$$

$$\therefore 0 = \frac{1}{2}e^0 + C$$

$$\therefore C = -\frac{1}{2}$$

$$\therefore t = \frac{1}{2}e^x - \frac{1}{2}$$

$$2t = e^x - 1$$

$$e^x = 2t + 1$$

$$x = \ln(2t + 1)$$

✓

✓

✓

4

$$\begin{aligned}
 (a) \quad (1+\sqrt{3})^4 &= 1^4 + 4 \cdot 1^3 \cdot \sqrt{3} + 6 \cdot 1^2 \cdot (\sqrt{3})^2 + 4 \cdot 1 \cdot (\sqrt{3})^3 + (\sqrt{3})^4 \\
 &= 1 + 4\sqrt{3} + 18 + 12\sqrt{3} + 9 \\
 &= 28 + 16\sqrt{3}
 \end{aligned}$$

✓ correct!
uses
Pascal or
binomial:

$$\therefore a = 28$$

$$b = 16$$

(Students who only write $28 + 16\sqrt{3}$ without $a=28, b=16$ get 2 marks)

✓ for $a=28$
✓ for $b=16$

(b) Coefficient of x^4 is nC_4 , coeff. of x^3 is nC_3

$$\therefore {}^nC_4 = 2 \cdot {}^nC_3$$

students who
write

$$2 {}^nC_4 = {}^nC_3$$

and $n=5$ get

only

1 mark.

(Answer not
checked)

$$\frac{n!}{(n-4)! 4!} = 2 \cdot \frac{n!}{(n-3)! 3!}$$

$$n-3 = 2 \cdot 4$$

$$n = 11$$

check: ${}^{11}C_4 = 2 \times {}^{11}C_3$. ✓ works

(c) The general term in $(a+b)^n$ is $T_{r+1} = {}^nC_r a^{n-r} b^r$

Thus general term in $(2x - \frac{1}{x^2})^{30}$

$$\text{equals } {}^{30}C_r (2x)^{30-r} \left(-\frac{1}{x^2}\right)^r$$

$$= {}^{30}C_r \cdot 2^{30-r} \cdot (-1)^r \cdot x^{30-r} \cdot x^{-2r}$$

$$= {}^{30}C_r \cdot 2^{30-r} \cdot (-1)^r \cdot x^{30-3r}$$

$$\text{We want } x^{12}, \therefore 30-3r = 12$$

$$\therefore r = 6$$

$$\therefore \text{coefficient desired is } {}^{30}C_6 \cdot 2^{24} = 9.96 \times 10^{24}$$

a correct
general
term

✓ (expand

or $r=7$
✓ Starting from

$${}^{30}C_{12} (2x)^{30-12} (-1)^{12} (x^{-2})^{12} \text{ gets no marks !!}$$

2nd method: 7th term from expansion and ${}^{30}C_6 2^{24}$

9

5 (a) $0.08 \text{ p.a.} \equiv 0.02 \text{ per quarter}$

(i) After 1 quarter:

$$A_1 = P + P(0.02) \\ = P(1.02)$$

After 2 quarters:

$$A_2 = A_1(1.02) + P(1.02) \\ = P(1.02)(1.02) + P(1.02) \\ = P(1.02^2 + 1.02), \text{ as required.}$$

* To get the marks the derivation (ie the process) must be shown

(ii) Similarly, after 20 years (= 80 quarters),

$$A_{80} = P(1.02^{80} + 1.02^{79} + \dots + 1.02^2 + 1.02) \\ = P(1.02^1 + 1.02^2 + \dots + 1.02^{80})$$

a G.P. with $a = 1.02$
 $r = 1.02$
 $n = 80$

$$\therefore S_{80} = \frac{1.02(1.02^{80} - 1)}{1.02 - 1} \\ = 51(1.02^{80} - 1)$$

So if we want $A_{80} = 1.000000$,

then $P = \frac{1\,000\,000}{51(1.02^{80} - 1)}$

$\therefore P = \$5\,059.52 \text{ (nearest cent).}$

5-(b)

$$(i) A_1 = 250\,000(1.005) - M$$

$$A_2 = A_1(1.005) - M$$

$$= [250\,000(1.005) - M](1.005) - M$$

$$= 250\,000(1.005)^2 - M(1 + 1.005)$$

$$\vdots$$

$$\therefore A_{60} = 250\,000(1.005)^{60} - M(1 + 1.005 + \dots + 1.005^{59})$$

as required

✓✓ must show process

$$(ii) 15 \text{ years} = 180 \text{ months}, \therefore A_{180} = 0$$

$$\therefore 250\,000(1.005)^{180} - M(1 + 1.005 + \dots + 1.005^{179}) = 0$$

✓

a G.P. with $a = 1$, $r = 1.005$
 $n = 180$

$$\therefore S_{180} = \frac{1(1.005^{180} - 1)}{1.005 - 1}$$

$$\therefore M = 250\,000(1.005)^{180} \times \frac{0.005}{1.005^{180} - 1}$$

$$= \$2\,109.64 \text{ (nearest cent)}$$

✓

$$(iii) \text{ Amount still owing after 5 years}$$

$$= A_{60} = 250\,000(1.005)^{60} - 2\,109.64 \left[\frac{1.005^{60} - 1}{1.005 - 1} \right]$$

✓

$$= \$190\,022.89$$

✓

5(b) continued

(iv) 7.2% p.a. = 0.6% per month.

We must pay off \$190 022.89 given

$r = 0.006$, $M = \$2400$:-

Suppose it is paid off after n months, then

$$190\,022.89 (1.006)^n - 2400 \frac{(1.006^n - 1)}{0.006} = 0$$

✓ correct
set-up &
start

$$\therefore 190\,022.89 (1.006)^n - 400\,000 (1.006^n - 1) = 0$$

$$\begin{aligned} \therefore 1.006^n &= \frac{400\,000}{400\,000 - 190\,022.89} \\ &= \frac{400\,000}{209\,977.11} \end{aligned}$$

✓ correct
process

$$n \log_{10} 1.006 = \log_{10} \frac{400\,000}{209\,977.11}$$

$$n = \frac{\log_{10} \left(\frac{400\,000}{209\,977.11} \right)}{\log_{10} 1.006}$$

$$\approx 107.7$$

ie. 108 months are required.

✓ correct
answer
(accept)

6

(a) The coefficient of x^6 in $(1+x)^{12}$ is ${}^{12}C_6 = \binom{12}{6}$

The coefficient of x^6 in $(1+x)^6 \cdot (1+x)^6$ is

$${}^6C_0 \cdot {}^6C_6 + {}^6C_1 \cdot {}^6C_5 + {}^6C_2 \cdot {}^6C_4 + {}^6C_3 \cdot {}^6C_3 \\ + {}^6C_4 \cdot {}^6C_2 + {}^6C_5 \cdot {}^6C_1 + {}^6C_6 \cdot {}^6C_0$$

$$= \binom{6}{0}^2 + \binom{6}{1}^2 + \binom{6}{2}^2 + \binom{6}{3}^2 \\ + \binom{6}{4}^2 + \binom{6}{5}^2 + \binom{6}{6}^2$$

$$[\text{since } {}^6C_k = {}^6C_{6-k}]$$

$$= \sum_{k=0}^6 \binom{6}{k}^2$$

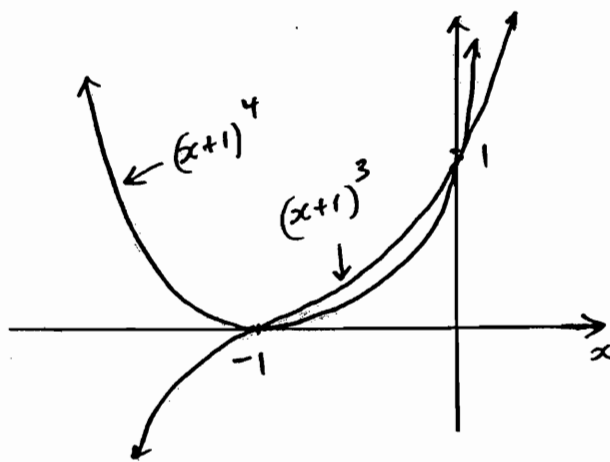
$$\text{Hence } \sum_{k=0}^6 \binom{6}{k}^2 = \binom{12}{6}$$

$$(b) \quad x^4 + 4x^3 + 6x^2 + 4x + 1 \leq x^3 + 3x^2 + 3x + 1$$

$$\therefore (1+x)^4 \leq (1+x)^3$$

$$\text{i.e. } (x+1)^4 \leq (x+1)^3$$

$$\therefore -1 \leq x \leq 0$$



✓ uses
pascal
relation
✓

✓ correct
start

✓ correct
answer